Semester 1 Examinations 2015/2016

Exam Code(s)	
Exam(s)	Higher Diploma in Software Design and Development
Module Code(s)	CT537
Module(s)	Software Engineering Methods
Paper No.	1
Repeat Paper	
External Examiner(s)	Prof. Liam Maguire
Internal Examiner(s)	Prof. Gerard Lyons Dr. Michael Madden Ms. Karen Young*
Instructions:	Candidates should answer **Question 1** and any **two other** questions. All questions carry equal marks.
Duration	2 hours
No. of Pages	5
Requirements: None	Release to Library: Yes ☐ No ☐

1. Draw a **context diagram**, a **systems level DFD** and one **third level DFD** to represent the data and processing involved in the following system description for a *University Library*.

> *University staff and students are usual users of the library. The library system records the books and journals owned by the library and records loans to members. Each member has a unique ID number. In order to borrow a book, the library member must show a valid ID card, which is checked to ensure that it is still valid against the student, staff or guest databases maintained by the university. The system must also check to ensure that the borrower does not have any overdue books or unpaid fines before he or she can borrow another book.*
>
> *The library maintains a large collection of borrowable items (e.g. books, videos, etc.) which it continuously updates both automatically when new editions become available and resulting from staff requests. There are reserved items that can only be borrowed by staff for one day. There are also seven days and 28 days loan items that can be borrowed by both staff and students. Sometimes books are lost or are returned in damaged condition. The manager must then remove them from the database and will sometimes impose a fine on the borrower.*
>
> *Staff members can borrow up to 8 items at a time, and students can borrow a maximum of 3 items. Librarians run the library and provide services (e.g. managing membership, lending items) to the members. Each librarian has a unique login name and has to login to the system before using it. Every Monday, the library contacts (by e-mail or text message) those people with overdue books. If a book is overdue by more than two weeks, a fine will be imposed and a librarian will telephone the borrower to remind him or her to return the book(s).*

(15)

(b) Provide a brief report detailing how the **implementation** of the *University Library* system in Q.1 will be approached. The report should detail the design method, programming language and framework employed, as well as the implementation strategy stating all assumptions made.

(5)

2. (a) "*Furious activity is no substitute for understanding*" H.H. Williams
What software engineering practices do you believe support the developer in gaining an understanding of the problem domain? In your experience, are these practices adequate? Give reasons to support your answer.

(8)

(b) You are the team lead of a systems design team that is implementing a multiple module project using new technologies and you have the responsibility of eliciting **requirements** for a customer who tells you he is too busy to meet you. Write a memo to this customer explaining the importance of a correct understanding of his requirements to project success.

(6)

(c) Software Design is both a process and a product. What is a software design? How does a software design differ from a software program? How is the quality of a software design assessed?

(6)

3. (a) Experienced software developers often say, "*Testing never ends, it just gets transferred from you [the software engineer] to your customer.*"

Assume that you are developing an online pharmacy that caters to senior citizens. The pharmacy provides typical functions, but also maintains a record for each customer so that it can provide drug information and warn of potential drug interactions.
Outline a ***test strategy*** for this system that will ensure comprehensive testing of the system before "customer tests" begin.

(8)

(b) Construct a structured specification for the supplier selection process of XYZ Ltd. described below. Explain why you chose the particular specification approach you used.

When I get a purchase order, my job is to pick a supplier from our file of available suppliers. Of course, some suppliers get eliminated right away because their price is too high, others because they've been temporarily "blacklisted" because of poor quality. But the real work that I do is to pick the optimal supplier from those that qualify -- the one that will deliver our order in the shortest amount of time. To do this I look at the supplier's location (<100 KM), the quantity of items being ordered (<100 items or greater than 100 items), and the date that we need the goods (within a week, within a month or longer), and I know which supplier will be the best one."

(8)

(c) What is a spike solution in XP? Describe the XP concepts of refactoring and pair programming.

(4)

4. (a) Risk identification is the first of a four-step approach to analysing risks in development efforts. What are the other three steps?

(2)

You are the project manager for a major software company. You've been asked to lead a team that's developing "next generation" word-processing software. Create a *risk table* for the project.
Develop a *risk mitigation strategy* and specific risk mitigation activities for three of the risks noted.

(6)

(b) Describe the concept of *software process maturity* in your own words. What different SPI (Software Process Improvement) frameworks are you aware of in use today?
What are the key organisational characteristics needed to successfully improve an organisation's software process?

(6)

(c) (i) What is the *Software Process*? Distinguish between the *Prototyping* and *Waterfall* approaches to the Software Process.

(2)

(**ii**) What types of software projects are amenable to the *Prototyping* model? Provide two examples.

(2)

(**iii**) What types of software projects are amenable to the *Waterfall* model? Provide two examples.

(2)